



National Aeronautics and
Space Administration

Intro to the ASTRA Initiative

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The Astrophysics Strategic Technology & Research Accelerator (ASTRA) Initiative

2

ASTRA is the superhighway to cosmic discovery. A multi-mission incubator that accelerates the time from innovation to exploration, by ensuring that there is always an astrophysics strategic mission at the ready.

ASTRA is an initiative that:

- Matures multiple strategic astrophysics missions at different levels of readiness, to be ready to fly when appropriate.
- Embraces Astro2020 recommendations, emerging industry capabilities, and strategic NASA documents.
- Ensures continuous, programmatically responsible, groundbreaking astrophysics discovery.

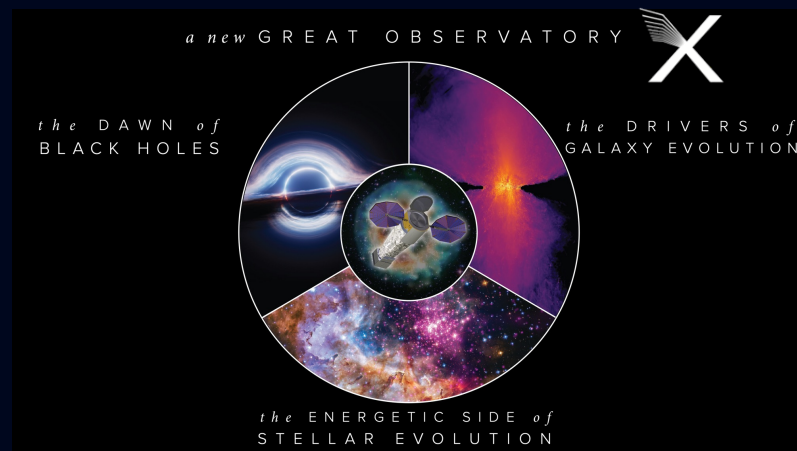
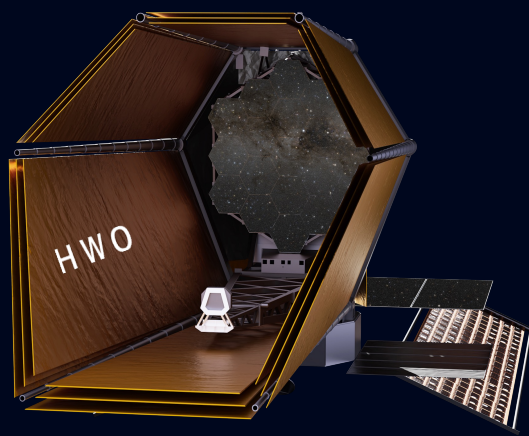
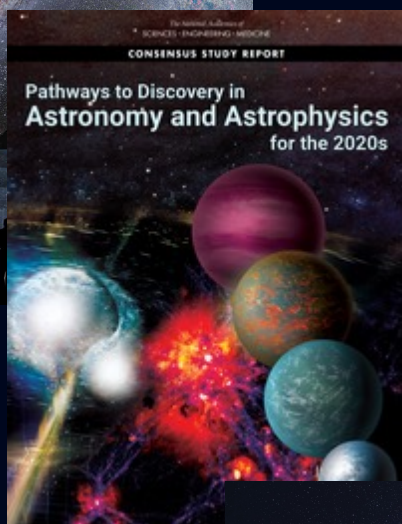
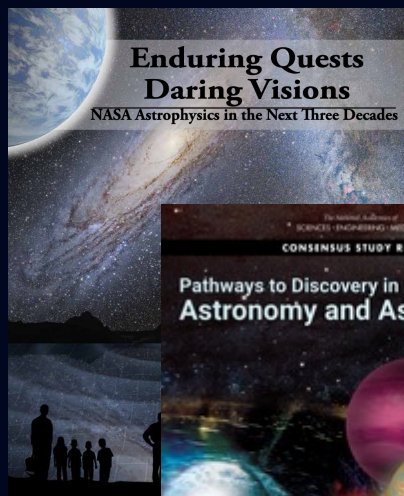
Celeriter ad ASTRA!



Westerlund 2 is a multi-wavelength view of a cluster of young stars located ~20,000 light years from Earth. Credit: X-ray: NASA/CXC/SAO/Sejong Univ./Hur et al.; Optical: NASA/STScI

Looking Forward

3



APD is asking the community to work with the PAGs to provide a *small* set of (4-6) candidate large strategic (>\$1B) mission concepts to study as entrants into the ASTRA Initiative.

APD is encouraging traditional & non-traditional architectures, that may include:

- Single observatories that support generations (e.g., Hubble, Chandra, JWST, HWO)
- Constellations of suites of smaller missions that collectively provides Flagship-scale science
- Space-based (and lunar) radio mission concepts

Selected mission concepts are in addition to:

- X-Ray flagship study (which is ramping up now)
- Far-IR flagship study

ASTRA PROCESS

5

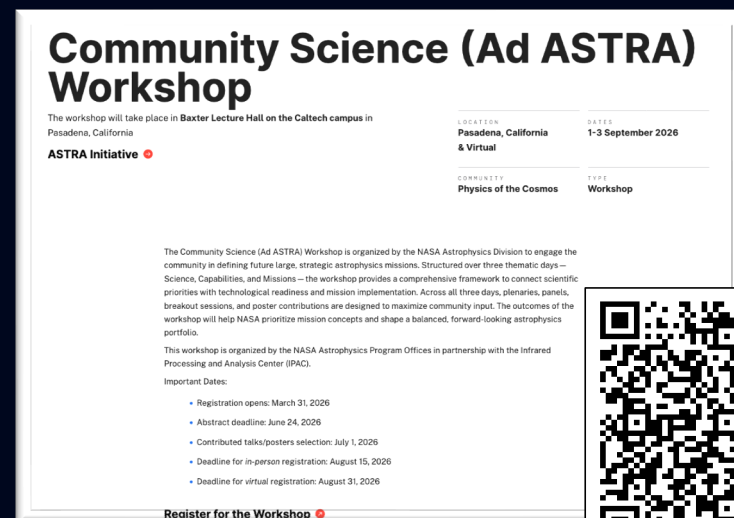
APD is encouraging the community to consider the following when prioritizing concepts for study:

- Industry and commercial capabilities, international partners, public/private partnerships, and the current and planned international fleet of missions.
- NASA's foundational strategic documents.
- Technology advancement needs over the next decade.

How to Participate:

- Through the PAGs!
 - Cosmic Origins (COPAG),
 - Exoplanet Exploration (ExoPAG)
 - Physics of the Cosmos (PhysPAG)

Each PAG may solicit the astronomy and astrophysics community for mission concept ideas. To accomplish this, each PAG is empowered to envision and use its own process.



Community Science (Ad ASTRA) Workshop

The workshop will take place in **Baxter Lecture Hall** on the Caltech campus in Pasadena, California

ASTRA Initiative

LOCATION Pasadena, California & Virtual	DATES 1-3 September 2026
COMMUNITY Physics of the Cosmos	TYPE Workshop

The Community Science (Ad ASTRA) Workshop is organized by the NASA Astrophysics Division to engage the community in defining future large, strategic astrophysics missions. Structured over three thematic days – Science, Capabilities, and Missions – the workshop provides a comprehensive framework to connect scientific priorities with technological readiness and mission implementation. Across all three days, plenaries, panels, breakout sessions, and poster contributions are designed to maximize community input. The outcomes of the workshop will help NASA prioritize mission concepts and shape a balanced, forward-looking astrophysics portfolio.

This workshop is organized by the NASA Astrophysics Program Offices in partnership with the Infrared Processing and Analysis Center (IPAC).

Important Dates:

- Registration opens: March 31, 2026
- Abstract deadline: June 24, 2026
- Contributed talks/posters selection: July 1, 2026
- Deadline for in-person registration: August 15, 2026
- Deadline for virtual registration: August 31, 2026

Register for the Workshop



ASTRA TIMELINE

6

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|----------------|---|
| May 2026 | Workshop on Innovation for Astrophysics Missions. A presentation will be provided to the community prior to the Community Science Ad ASTRA Workshop. |
| June 2026 | 248th AAS Meeting. An overview of the ASTRA Initiative will be presented in a Special Session, invited talks from the ADP Division Director and others, followed by a panel Q&A session. |
| September 2026 | NASA Community Science Ad ASTRA Workshop. A presentation will be provided to the community by the end of the year. |
| December 2026 | Joint PAG Virtual Presentation. A public presentation will be given by PAG representatives to provide community input to the NASA Astrophysics Division Director regarding which mission concepts to study*. |
| January 2027 | 249th AAS Meeting. Mission concept study selections will be announced by the NASA Astrophysics Division Director during the Town Hall. |

***Study process details will be provided at later date.**

ASTRA Initiative

The Future of NASA Astrophysics



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Astrophysics Strategic
Technology & Research
Accelerator (ASTRA) Initiative.

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Cosmic Origins Science

The Astrophysics Strategic Technology & Research Accelerator (ASTRA) Initiative

To maintain US leadership, NASA's Astrophysics Division (APD) is launching the Astrophysics Strategic Technology and Research Accelerator (ASTRA) Initiative.

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About the ASTRA Initiative

How does the universe work? How did we get here? Are we alone?

At every stage, NASA will collaborate with industry, academia, and international partners to advance necessary technologies, manufacturing capabilities, and reduce costs.

ASTRA aims to reduce the total cost, time-to-science, and schedule risk of future strategic missions, aligning with the guidance from Astro2020, LSSM, and AMP.



This is an artist's impression of how the very early universe (less than 1 billion years old) might have looked when it went through a voracious onset of star formation, converting primordial hydrogen into myriad stars at an unprecedented rate. Back then the sky would have looked markedly different from the sea of quiescent galaxies around us today. The sky is ablaze with primeval starburst galaxies; giant elliptical and spiral galaxies have yet to form. Unlike today there is very little dust in these galaxies, because the heavier elements have not yet been cooked up through nucleosynthesis in stars. Recent analysis of Hubble Space Telescope deep sky images supports the theory that the first stars in the universe appeared in an abrupt eruption of star formation, rather than at a gradual pace.

Credit: NASA, STScI A. Schaller



ASTRA
QUESTIONS?

